DRISH SHAH

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Available full time for Summer 2027

EDUCATION

Purdue University, West Lafayette — Expected December 2028

B.S. Robotics Engineering Technology — *GPA 3.91 / 4.00, Dean's List*

- Coursework: Statics, Dynamics, Machine Design, CAM and CNC Machining, Instrumentation and Data Acquisition, Industrial Controls

EXPERIENCE

Mechanical Engineering Intern, Robotics — May 2026 to Present

Vibe Robotics, humanoid robotics startup — *West Lafayette, IN*

- Design hardware for the Vibe 01 home humanoid across the drive system, chassis, and actuation, from CAD to printed prototype to assembly on the robot; wire and integrate actuator motors.
- Developed the robot's dexterous hand and a laundry gripper for an October 2026 hotel pilot, tuning TPU finger geometry and compliance with fingers tapered palm to tip and relief windows that shrink toward the tip.
- Scoped a modular end effector system for autonomous tool swapping across a mop, plate gripper, laundry folder, and general gripper on one mechanical interface.
- Iterated through design reviews with the electrical and software teams; benchmarked leading home robot hands.

Undergraduate Researcher — August 2025 to Present

TRACE Lab, Prof. Yan Gu, Office of Naval Research humanoid program — *West Lafayette, IN*

- Designed the rail grasping gripper for Digit on the Bus, the program's land based stand in for shipboard motion, so the Digit humanoid holds on through 600 N vibration spikes inside a 150 x 150 x 120 mm volume budget.
- Modeled it in SolidWorks around a DM-J4310 actuator and 32 tooth transmission, with a carbon fiber frame and Fin Ray TPU fingers sized for 32 to 38 mm rails; ran FEA on the structure.
- Load tested version one under static rail engagement; PLA stiffness limits found in testing drove a redesign.
- Built and integrated hardware for field trials aboard a vessel in Norfolk, VA, including wiring and control of actuator motors.
- Starting end effector work on a TRACE Lab and Eli Lilly collaboration: Spot tooling for autonomous gas leak inspections.

PROJECTS

Modular Humanoid Hand and Arm | *Mechanical team, Purdue Humanoid Robot Club* — 2025 to 2026

- Built a 6 DOF arm with a 3 DOF wrist and a modular hand, replacing tendons with linkages so each finger runs on one actuator.
- Ran tolerance analysis for printed parts, fastener selection, and BOM creation, then assembled the arm and tested range of motion.

Architected Origami Metamaterials | *Independent research* — 2026

- Measured Poisson's ratios from −0.6 to +0.57 across five tessellations on a custom EV3 test rig with dual dynamometers; presenting author at the Society of Engineering Science Annual Meeting, October 2026.

LILY: Autonomous Water Quality Monitor | *Purdue TECH 120* — 2025

- Designed a floating monitor in Onshape across two prototypes, with a sealed, waterproof electronics enclosure and a Raspberry Pi logging temperature, ORP, and TDS; field validated on a frozen lake at 95% accuracy against professional grade systems priced from $219 to $499.

LEADERSHIP AND SERVICE

Co Captain and Mentor, FRC Sigma 9692 and FTC Sigma 20890 — 2021 to 2025

- Led 26 member FRC and 12 member FTC teams through full robot development cycles, from concept to competition.
- Raised $52,000, taught 3,000 students, and established 27 robotics labs across underserved Indian schools.
- Purdue roles: Student Ambassador, School of Engineering Technology (1 of 14); Building Manager, Purdue Memorial Union and Stewart Center; Techie Tutor (2026 to Present).

SKILLS

CAD and PDM: SolidWorks, Siemens NX, CATIA V6, PTC Creo, Onshape, Fusion 360

Analysis and Test: FEA, tolerance analysis, load and vibration testing, MATLAB, MuJoCo, Isaac Sim

Manufacturing: DFM and DFA, GD&T (ASME Y14.5), CNC machining and CAM, sheet metal parts, fastener and material selection, BOM and vendor sourcing

Fabrication: FDM 3D printing in PLA and TPU, laser cutting, hand machining

Electrical and Programming: PCB design, motor wiring and control, Raspberry Pi, Arduino, DAQ (NI USB 6008), Python, C/C++

AWARDS

Presented with Distinction: Purdue Office of Undergraduate Research Conference, 2025: humanoid hand research

Engineering Inspiration Award: FIRST Robotics Competition, 2025: second highest honor in the competition

Inspire Award: FIRST Tech Challenge Asia Pacific Championship, 2023: highest honor awarded